Sprint 2 User Use Cases

Detailed User Creator and Reviewer journeys with honest acceptance states

IRON CHEF v3 | Vo Trung Tin | Reviewed 11 October 2026

These use cases specify intended behavior at the Member–Lead boundary. A screen or adapter can be implemented while a model, reviewed safety policy, official graph, or research export remains pending. The state of each journey is therefore recorded alongside its expected flow.

Source baseline and interpretation

Source	Evidence used
Sprint 2 report supplied by the user	Work assignment: Vo Trung Tin owns UI requirements, use cases, data flows, and Member scaffold acceptance. The sprint runs 23 September–10 October 2026.
Member backlog, branch codex/member-sprint12-deliverables	Epic 1–12 implementation states and open gates G1–G10; engineering checkpoint dated 6 October 2026.
Lead backlog v109, origin/master 8bb8167	As of 9 October 2026, Stage 3 is artifact-verified through exact step 327/561. Segment 6 was launched toward step 374. Final model quality and scoring remain unverified.
Browser captures dated 6 October 2026	Actual frontend views of /explore/graph and /scan, with the graph backed by local seed data.
Frontend and monolith code	Routes, request boundaries, error states, fields, and adapter behavior reviewed for this edition.

Evidence rule: a browser image proves that a screen appeared during that run; code shows an implemented path; a backlog records the team's checkpoint. None alone proves production deployment, model quality, clinical safety, or a research result.

Use case index

Case	Journey and current boundary
UC-01	Creator authors a recipe; saved ingredient context still needs cross-service verification.
UC-02	User scans or uploads ingredients; editing detections and accepted models remain open.
UC-03	User requests a safe substitution; reviewed production policy remains open.
UC-04	User explores the graph; UI works with seed data, official import remains open.
UC-05	User records cooking feedback; full outcome and recovery evidence remains open.

Case	Journey and current boundary
UC-06	Reviewer inspects evaluation; dashboard works, verified Lead exports remain open.
UC-07	Reviewer tracks thesis evidence; 14-chapter mapping exists, captures remain open.

UC-01 Creator authors a recipe

Field	Description
Actors	Authenticated Creator
Precondition	The Creator may create or edit a recipe.
Trigger	Select Create or Edit.
Outcome	A saved or published recipe retains ingredients and quantities.
Traceability	UI-01, UI-10; Member product baseline

Main flow

1. Enter title, ingredients, quantities, cooking steps, and media.
2. Save a draft; the system validates fields and preserves the ingredient list.
3. Preview, revise, and publish the recipe.
4. Provide stored recipe ingredients as context when a User asks for a substitution.

Alternatives and failures

- On invalid fields or failed save, show the error and retain the draft input.
- On insufficient permissions, reject the edit or publication action.

Acceptance: Reload preserves ingredient identities and amounts; the substitution request receives the correct recipe context. Community engagement is not recorded as a research label.

UC-02 User scans and reviews ingredients

Field	Description
Actors	User, browser, configured detector
Precondition	The /scan route opens; image submission has a valid caller identity.
Trigger	Choose camera or upload.
Outcome	The reviewed list proceeds, or the journey stops with a clear state.
Traceability	UI-02; G8; Sprint 2 sections 2.2–2.4

Main flow

1. Request camera permission only after an explicit action, or open image selection.
2. Validate non-empty image type and size before same-origin submission.
3. If an accepted detector responds, show name, location, confidence, and actual source.
4. Let the User correct or remove detections before recipe matching or substitution.
5. Continue only with a reviewed ingredient list.

Alternatives and failures

- Permission denied or camera unsupported: show the reason and offer upload or manual entry.
- Unconfigured detector: return 503 integration pending; never invent live detections.
- Invalid, empty, or >10 MiB image: return 400/413 without calling a model.
- Open UI gap: the observed scan page has no per-detection edit control; that acceptance step still needs implementation and verification.

Acceptance: Browser evidence covers controls only. Verify permission refusal, camera switching, stream cleanup, mobile devices, and a Lead-accepted detector.

UC-03 User requests a safety-aware substitution

Field	Description
Actors	User, AI service, reviewed policy provider
Precondition	An ingredient and context exist; an empty profile never defaults to Safe.
Trigger	Select an ingredient and reason for substitution.
Outcome	The User chooses or declines with safety state and limitations visible.
Traceability	UI-03–06, UI-09; G3/G4/G7

Main flow

1. Collect ingredient, reason, recipe context, dietary tags, saved allergen flags, and session identity.
2. Request candidates and preserve their receipt and evidence fields when returned.
3. Resolve local profile conflicts and source policy into Safe, Check, or Blocked.
4. Display candidate identity, quantity ratio, confidence, source, and supported compound or nutrition evidence. Show missing measurements as unavailable.
5. Allow selection of a non-Blocked candidate or decline all candidates; record real receipts when available.

Alternatives and failures

- Missing, malformed, or uncertain policy yields Check; absence of an allergen name does not imply Safe.
- A conflict yields Blocked and disables the primary action.
- Service failure is visible and never replaced by unlabeled demo output.
- Missing compound or nutrition measurements are not displayed as measured zero.

Acceptance: Test authenticated profile → policy → card end to end, including custom and empty profiles, correct sources and receipts, and a reviewed Lead policy.

UC-04 User explores ingredient relationships

Field	Description
Actors	Guest or User, Knowledge API
Precondition	The graph API returns a seed or an accepted imported collection and reports coverage.
Trigger	Open /explore/graph or search an ingredient.
Outcome	The User understands the returned relationships and partial coverage.
Traceability	UI-07; Epic 1/10; G5

Main flow

1. Load a bounded graph response and show its node, edge, and coverage counts.
2. Search names or aliases and center the selected result.
3. Drag nodes, pan or zoom, and select nodes or edges for details.
4. Show relationship type, context, ratio, allergen flags, and auxiliary evidence only when supplied.

Alternatives and failures

- No links: show zero rather than generate edges.
- Missing FooDB or USDA data: display unavailable or pending; missing allergen flags do not imply Safe.
- The seed API emits substitution only, even though other filter labels appear in the UI.
- API failure: show an actionable error and retry state.

Acceptance: The observed seed has 25 nodes and four edges; searching Ginger centers an isolated result. The official Lead graph import remains gate G5.

UC-05 User records cooking feedback

Field	Description
Actors	User, frontend, monolith, AI ingestion
Precondition	A cooking session and valid candidate receipt exist where the event is candidate-specific.
Trigger	Accept, reject, skip, or finish a cooking session.
Outcome	The UI accurately reports whether the feedback was stored.
Traceability	UI-09–10; G7

Main flow

1. Create a choice or post-session outcome event.
2. Send session ID, candidate receipt, and event identity to the backend.

3. Persist the event durably and project it through the agreed ingestion path.

4. Show success or an actionable retry state.

Alternatives and failures

- Never fabricate a missing receipt.
- Retry or replay must not count an event twice, and a failed projection must not be reported as successful learning.
- Likes, XP, and streaks are product interactions, not held-out evaluation labels.

Acceptance: Exercise accept/reject/skip/taste across services, retries, idempotency, failure and recovery. Do not claim model improvement without representative outcomes.

UC-06 Reviewer inspects sourced evaluation

Field	Description
Actors	Authorized Reviewer or Admin
Precondition	The reviewer is authenticated for /admin/evaluation.
Trigger	Open the dashboard or export a figure.
Outcome	The reviewer sees a truthful dashboard or export.
Traceability	UI-08; G1/G6

Main flow

1. Load bundled or configured remote Lead manifest data.

2. Validate structure, states, metrics, metadata, and internal consistency.

3. Show complete, pending, and rejected records honestly; hide pending numeric values and preserve negative findings.

4. Export charts with the provenance and limitations actually supplied.

Alternatives and failures

- Unauthenticated visitor is denied or redirected to sign-in.
- Malformed manifest or failed remote source fails closed without old scores.
- Different datasets or protocols are not presented as a matched winner comparison; a displayed hash does not verify an artifact.

Acceptance: Verify actual Lead artifacts, dataset split, protocol, seeds, strata, intervals, abstention, and review before treating a score as complete.

UC-07 Reviewer tracks thesis evidence

Field	Description
Actors	Reviewer, Member, Lead
Precondition	The /thesis workspace maps the 14 Lead chapters.
Trigger	Open a chapter evidence item.
Outcome	Each item has an honest state and traceable source.
Traceability	UI-11; G2/G9

Main flow

1. Open the chapter and unique artifact ID.
2. Read source route and pending, ready-to-capture, captured, or reviewed state.
3. Capture the real source when available; record caption, date, environment, and version.
4. Mark reviewed only after examining the evidence and claim limits.

Alternatives and failures

- Missing or unaccepted source remains pending.
- Seed/demo screenshots or planned deployment figures do not become production evidence.

Acceptance: Every submitted figure has source, date, version, access boundary, and reviewer. No chapter is marked finished from a plan alone.